Kubernetes Namespaces – Quick Revision

Why do we use Namespaces?

Used to separate Dev, Test, UAT and Production environments. Resources are isolated between namespaces.

1. List all namespaces

kubectl get ns
Displays all namespaces. Default namespaces: default, kube-system, kube-public, kube-node-lease.

2. List pods in all namespaces

kubectl get pods -A
or **kubectl get pods --all-namespaces**
Shows pods from every namespace.

3. View system pods

kubectl get pods -n kube-system
Shows Kubernetes system components like CoreDNS, kube-proxy, Metrics Server.

Default namespace

If no namespace is specified, resources are created in the **default** namespace.

4. Create a namespace

kubectl create ns dev
or **kubectl create namespace dev**

5. Specify namespace in YAML

Add **metadata.namespace: dev** in the manifest to always create the resource in that namespace.

6. Create resource in a namespace

kubectl create -f pod.yaml -n dev
If YAML has no namespace, this creates it in dev.

7. Check current namespace (kubens)

kubens – Shows current namespace.
kubens test – Switches to test namespace.

8. Check current namespace (kubectl)

kubectl config view --minify | grep namespace
Displays current namespace.

9. Change default namespace

kubectl config set-context --current --namespace=dev
Changes the current context's default namespace.

Important Interview Commands

kubectl get ns – List namespaces
kubectl get pods -A – List all pods
kubectl get pods -n kube-system – System pods
kubectl create ns dev – Create namespace
kubectl create -f pod.yaml -n dev – Create resource in namespace
kubectl config set-context --current --namespace=dev – Change default namespace
kubectl config view --minify | grep namespace – Check current namespace
kubens – Show current namespace
kubens test – Switch namespace

Interview Summary

A namespace is a logical partition in Kubernetes used to isolate and organize resources within the same cluster.